

Explanation

There are three functions defined in the driver. Instead of calling them directly, we pass each into a newly created ‘sthread’. All threads will use the same virtual memory space, meaning no new processes will be forked. Context switching will still happen, but only between Thread Control Blocks. This requires much less overhead than switching between entire Process Control Blocks.

Address space for our threads will look like this:

max	OS Memory	
	Stack func1	Base Pointer Bothell Thread
		Stack Pointer Bothell Thread
	Stack func2	BP Seattle Thread
		SP Seattle Thread
	Stack func3	BP Tacoma Thread
		SP Tacoma Thread
0	Heap (malloc depends on this, grows up)	
	data	
	text	Program Counter Tacoma Thread
		PC Seattle Thread
		PC Bothell Thread

If we called all three functions instead of creating threads, there would be three structs like the one above, rather than one struct that contains three threads.

Schedule

These are the user-defined threads and their values of *i*:

■ Bothell Thread (func1) ■ Seattle Thread (func2) ■ Tacoma Thread (func3)

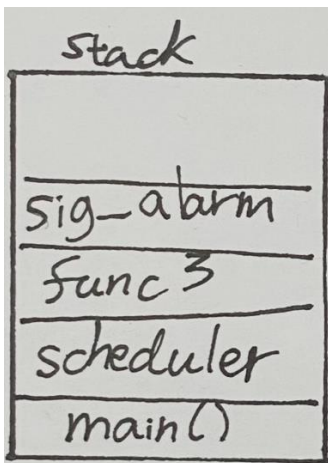
0	1	2	3	4										5	6	7	8	9												
					0	1	2	3	4											5	6	7	8	9						
										0	1	2	3	4												5	6	7	8	9

Output

```
[hreyes@csslab9 four-twenty-four]$ ./a.out
scheduler: initialized
func1: Bothell 0
func1: Bothell 1
func1: Bothell 2
func1: Bothell 3
func1: Bothell 4
func2: Seattle 0
func2: Seattle 1
func2: Seattle 2
func2: Seattle 3
func2: Seattle 4
func3: Tacoma 0
func3: Tacoma 1
func3: Tacoma 2
func3: Tacoma 3
func3: Tacoma 4
func1: Bothell 5
func1: Bothell 6
func1: Bothell 7
func1: Bothell 8
func1: Bothell 9
func2: Seattle 5
func2: Seattle 6
func2: Seattle 7
func2: Seattle 8
func2: Seattle 9
func3: Tacoma 5
func3: Tacoma 6
func3: Tacoma 7
func3: Tacoma 8
func3: Tacoma 9
scheduler: no more threads to schedule
```

Stack Layers

Stack layers “when a timer interrupt (=sig_alarm) was just asserted while func3 is running its for loop”.



Sources

- Fukuda, Munehiro. “Program 2A Hints.” CSS 430: Operating Systems, 2026.
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- <https://cs61.seas.harvard.edu/wiki/2017/Section12/>
- Operating Systems 10th Edition